Recorded wind learning and monitoring
Dashed: incomplete recording · ×: physical contact · no extrapolation

Scenario 1

Scenario 2

Whole-map wind RMSE (m/s)

Mean clarity in Region A

Mean clarity in Region B

Accumulated clarity deficit

Executed time (s)

Executed time (s)

Baseline 1: learn everywhere
Clarity-weighted learning
Clarity weighting + route handoff to MBD
MBD with wind uncertainty checks
MBD with targeted wind learning
Lower-cost opportunity learning
Measured-map MBD (unrestricted initial data)